Complete System Features, User Roles & Architecture Specification

Institution: Nitte Deemed to be University • Mangaluru / Nitte Campus
Regulatory Directives: CPCB E-Waste Management Rules 2022 • NAAC Criterion 7.1.3 • GEM 2024
Production Edge Web App: https://ecoloop-campus.vercel.app • **GitHub:** github.com/ibaibhav0/ecoloop

1. Executive Summary & Mission Architecture

EcoLoop is a mission-grade, closed-loop institutional circularity and e-waste tracking platform deployed exclusively for **Nitte Deemed to be University**. It transforms ad-hoc university electronics disposal into a verified, cryptographic chain-of-custody connecting students, campus residential collection teams, administrative authorities, and state-authorized Central Pollution Control Board (CPCB) recyclers.

Strict Real-Data Mandate: Unlike mock prototypes, EcoLoop operates with 100% adherence to real data. When zero items exist in the database, all telemetry counters, charts, diversion rates, and hostel standings default to zero. Every statistic reflects verifiable student handovers and calibrated scale readings.

100% Zero Landfill Guaranteed closed-loop CPCB authorized dismantling	**Instant Food Coupons** Tiered campus meals for donated hardware	**1.44 kg CO2e / kg** GEM 2024 & US EPA carbon avoidance ratio	**10 PTS / kg** Academic green credits for student participation

2. Complete Web Application Features (14 Core Capabilities)

Feature 1: SpaceX-Inspired Industrial Dark Telemetry Aesthetic

- **Austere Mission-Control UI:** Deep dark canvas (#000000) with monospaced data readouts, high-contrast borders (#3a3a3f), and macro photography of circuit boards and compute server hardware.
- **Dynamic Mission Carousel:** Rotates across 3 institutional mission profiles: Lab Hardware Decommissioning, Rare Earth Harvesting, and NAAC 7.1.3 Compliance Tracking.
- **Universal Navigation & Quick Sign-Out:** Single-tier navigation bar with dedicated **Sign Out** button on top right for instant account switching, plus universal ← **Back** navigation on all subpages.

Feature 2: Serialized Hardware Intake & Guaranteed QR Digital Tag Minting

- **Granular Category Ingestion:** Supports Laptops, Mobile Phones, Chargers & Cables, Lithium Batteries, Circuit Boards, Monitors, Home Appliances, and Other Electronics.
- **Instant Cryptographic Tracking ID:** Generates permanent serial tags (NITTE-EWASTE-XXXX) stamped with user USN, hostel, mass, and category.
- **Guaranteed QR Passport Render:** Client and server fallback engine guarantees immediate rendering of the QR barcode for immediate on-screen inspection, printing, or download.

Feature 3: Instant Quality-Based Campus Food Court Voucher Engine

- **Live Hardware Quality Evaluation:** Upon physical handover, campus collection officers assess the condition of the electronic item across 3 standardized tiers (Grade A, Grade B, Grade C).
- **Automated Voucher Minting:** The server cryptographically mints a unique food voucher (e.g., NITTE-DELUXE-7X9B2) and immediately credits it to the student's account.
- **Multi-Tier Dining Rewards:**
  - **Grade A (Excellent / Working):** Free Deluxe Campus Meal Combo + Cold Beverage (■**180**) at Nitte Central Food Court & Nescafe Lounge.
  - **Grade B (Good / Minor Wear / Repairable):** Free Special Masala Dosa / Grilled Sandwich + Fresh Juice (■**100**) at Campus Cafeteria.
  - **Grade C (Scrap / Broken / End-of-Life):** Free Cold Coffee / Hot Chai + Samosa Refreshment (■**50**) at all Campus Food Kiosks.
- **Collector Handover Celebration Screen:** Instantly displays the issued voucher code, value, and food outlet to the collector and student upon collection.

Feature 4: Student Food Court Voucher Digital Wallet

- **Dedicated In-Dashboard Wallet:** Students have a prominent 'MY FOOD COURT VOUCHERS & MEAL COUPONS' section on their personal dashboard displaying all earned vouchers.
- **One-Click Counter Redemption:** Each voucher card displays the distinct voucher code, commercial value, authorized food outlet, and 30-day validity window.
- **Public Tracker Integration:** Food voucher status and verified quality grade are visible in the item's digital passport via the public status tracker.

Feature 5: 5-Stage Cryptographic Chain-of-Custody & Status Tracker

- **Finite Lifecycle State Machine:** REGISTERED → SCHEDULED → COLLECTED → HANDED_OVER → VERIFIED_RECYCLED.
- **Public Tracker Modal & Page (/tracker):** Search by Tracking ID or Email to inspect real-time milestone steppers, operator name, hostel, calibrated mass, and custody handovers.

Feature 6: Optical Camera QR Scanner & Photo Upload Decoder (/scan)

- **Live Camera Barcode Scanner:** High-speed browser optical video scanner for phone and laptop cameras.
- **Photo Upload & Manual Code Fallback:** Optical decoder decodes QR codes directly from gallery images, with direct alphanumeric lookup fallback.

Feature 7: Universal Responsive Bounded Dialog Architecture (Zero-Clip Modals)

- **Viewport-Constrained Scroller:** All modal dialogs (CollectWeightModal, RegisterItemModal, StatusTrackerModal, SchedulePickupModal, VerifyRecycledModal) enforce max-h-[92vh] flex flex-col.
- **Sticky Header & Native Scrolling:** Dialog headers remain pinned while form contents and radio cards scroll smoothly via overflow-y-auto on trackpads, mousewheels, and touch screens.

Feature 8: Calibrated Tare & Precision Mass Weigh-In

- **Scale-Verified Weigh-In:** Collectors enter precision mass (kg) at campus collection depots, automatically replacing student estimates with calibrated data.
- **Dynamic Point Multiplier:** Automatically credits **+10 Eco-Credits per kg** to the student's balance.

Feature 9: Interactive Campus Geospatial Vector Map (/map)

- **Nitte Campus Drop-Off Grid:** Visual map plotting collection points across residential hostels (Nandani, Kaveri, Kumardhara, Saupernikha, Shambhavi, Netravati) and Central Depot.
- **GPS Distance Calculation:** 1-click 'USE GPS' computes geodesic distance and walking minutes to the closest e-waste collection bin.

Feature 10: Carbon Offset Ledger & Formal Digital EPR Certificates (/carbon)

- **Peer-Reviewed Emission Modeling:** Calculates carbon avoided at **1.44 kg CO2e / kg** of diverted e-waste (GEM 2024 / US EPA standards).
- **Equivalence Translations:** Urban trees planted equivalent, gasoline km avoided, and smartphone recharges saved.
- **Formal EPR Certificates:** Generates verifiable statutory certificates with serial IDs (EPR-2026-XXXX) and QR stamps.

Feature 11: NAAC Criterion 7 Institutional ESG Report & PDF Exporter (/compliance, /report)

- **NAAC Metric 7.1.3 Compliance:** Tailored for university institutional quality and environmental audits.
- **Statutory Summary Grid:** Total diverted mass, zero-landfill yield, toxic chemical mitigation (Lead, Mercury, Cadmium).
- **1-Click Standalone PDF Export:** Exports formal university documentation with accreditation seals.

Feature 12: 30-Day CPCB Storage SLA Watchdog & Automated Resolution

- **Statutory Storage SLA Watchdog:** Enforces CPCB Rule 2022 forbidding bulk campus storage beyond 30 calendar days.
- **Color-Coded Aging Telemetry:** NORMAL (<23 days), WARNING (23–30 days), and CRITICAL BREACH (>30 days).

Feature 13: 50+ Item Hardware Catalog & Elemental Hazards Directory

- Comprehensive hardware taxonomy across 7 sectors with scientific toxic hazard breakdowns and recoverable rare earth elements.

Feature 14: Omni-Channel WhatsApp Cloud API Bot Companion

- Replicates 100% of web app operations over WhatsApp chat (registration, pickup scheduling, tracking, and food voucher lookups).

3. Hardware Quality & Food Court Voucher Valuation Matrix

Quality Grade	Hardware Condition Criteria	Instant Food Coupon Reward	Value	Redemption Outlets	Eco-Credits
Grade A **Excellent / Working**	Boots up, intact display/chassis, fully functional or reusable components (laptops, phones, tablets).	**Free Deluxe Hot Meal Combo** Complete burger/rice meal + 1x Cold Beverage.	**■180** **100% Free**	Nitte Central Food Court & Nescafe Lounge	+10 PTS / kg (Scale Mass)
Grade B **Good / Repairable**	Minor cosmetic damage, repairable circuit boards, functional sub-assemblies (cables, appliances).	**Free Special Masala Dosa / Grilled Sandwich** Includes 1x Fresh Fruit Juice.	**■100** **100% Free**	Campus Cafeteria & Fast Food Center	+10 PTS / kg (Scale Mass)
Grade C **Scrap / Recyclable**	Broken glass, burnt PCBs, dead batteries, end-of-life metals destined for smelting & recycling.	**Free Iced Coffee / Hot Masala Chai** Includes 2x Fresh Crispy Samosas.	**■50** **100% Free**	All Campus Food Kiosks & Cafeterias	+10 PTS / kg (Scale Mass)

4. Multi-Role Campus Access Control Matrix

Role Persona	Portal / Interface	Core Privileges & Operating Capabilities	Authentication & Scope
Student Contributor	Student Dashboard Mobile WhatsApp	• Register e-waste with estimated mass & hostel pickup slot • Download/print unique 2D QR Digital Hardware Tag • Access live tracking status, audit logs, and eco-credits • **Receive & redeem instant campus food court meal vouchers** • Download personal certified EPR Carbon Offset certificates	University Email (@nitte.edu.in) USN Verification

Campus Collector	Operations Queue Mobile Camera Scanner	• View pending pickup requests by hostel • Scan QR tags via smartphone camera or upload • Input calibrated physical scale weight (kg) • **Evaluate hardware quality (Grade A/B/C) to mint food coupons** • Attest custody transfer into central university depot	Role: collector Staff PIN verification
Authorised Recycler	Recycler Hub (/recycler) Destruction Console	• Verify inbound shipments transferred from campus depot • Log terminal dismantling & material recovery (Gold, Silicon, Copper) • Issue cryptographically signed Certificate of Safe Recycling	Role: recycler CPCB Reg. Verification
University Admin	Admin Console (/admin) Executive ERP	• High-altitude real-time telemetry (real data only) • Monitor 30-day storage SLA breaches and execute dispatch • Export NAAC 7.1.3 institutional sustainability audit reports • Track hostel standings and campus diversion yield	Role: admin IQAC Officer Credentials

5. Complete Web Pages & Application Routes Directory

Route	Module Name	Detailed Page Functionality & User Actions
/	**Mission Control Dashboard**	Hero circuit showcase, real-time campus telemetry counters, active item feeds, personal food voucher wallet, and rapid-action modals.
/register	**Standalone Registration**	Full-page hardware intake form with category selection, tare weight calculation, and instant QR digital passport generation.
/tracker	**Consignment Status Tracker**	Public telemetry search by tracking code or email with 5-stage progress steppers, handover logs, and food voucher status.
/scan	**Optical QR Barcode Scanner**	Multi-modal scanner supporting live webcam, camera photo upload, or direct tracking code lookup.
/map	**Campus Drop-Off Vector Map**	Interactive 2D vector coordinate map of hostel drop bins with real-time GPS geolocation and walking distance computation.
/carbon	**Carbon Credit & ESG Ledger**	Dynamic GHG emissions avoided modeling (1.44 kg CO2e / kg) and downloadable official EPR certificates.
/compliance	**Regulatory ESG Compliance**	Statutory CPCB Form-6 manifest management, 30-day storage SLA countdown watchdog, and recycler license monitors.
/report	**NAAC 7.1.3 Formal Audit Report**	Institutional environmental sustainability audit report with 1-click standalone PDF exporter.
/partners	**Authorized Recyclers Directory**	Directory of accredited, state-authorized CPCB e-waste recyclers with capacities, authorization IDs, and facility locations.
/developers	**Developer REST API Console**	Interactive developer sandbox with live endpoint testing, cURL / Node.js code snippets, and API key provisioning.
/story	**Campus Circular Story**	Institutional origin story, zero-landfill campus vision, and rare-earth recovery mission for Nitte Deemed to be University.
/admin	**Institutional Administration**	Executive operations dashboard for sustainability officers: aggregate mass, hostel rankings, and bulk recycler transfers.
/recycler	**Recycler Inbound & Certification**	Authorised recycler intake portal to confirm receipt, log recovered materials, and issue Safe Recycling Certificates.

6. Technology Architecture & Production Deployments

Layer	Technologies Employed	Capabilities & Deployment Specifications
Frontend Web App	**Next.js 16** (App Router, Turbopack) Tailwind CSS + Vanilla CSS Tokens Lucide React • Html5-Qrcode	High-performance responsive web app. SpaceX-inspired dark theme, sub-second route transitions, client-side QR generation, camera optical scanner, and bounded scrollable dialogs.
Backend REST API	**Node.js & Express** (ES Modules) Mongoose ORM • JWT Auth OpenAPI 3.0 • ReportLab Engine	Stateless REST API managing custody transitions, food coupon minting, 30-day SLA calculations, carbon offsets, and Form-6 manifest synthesis.
Database	**MongoDB Atlas** (Cloud Dedicated) Resilient In-Memory Fallback Store	High-availability document store with automated failover and zero-latency in-memory cache ensuring 100% uptime even during network partitions.
WhatsApp Engine	**Meta WhatsApp Cloud API v21.0** Node.js Webhook • Jimp + jsQR	Event-driven webhook engine decoding user photo QRs and sending back real-time notifications, tracking updates, and food vouchers.
Hosting & Edge	**Vercel Edge Network** (Client) **Render Cloud** (Bot/API) GitHub (main branch)	• Production URL: **https://ecoloop-campus.vercel.app** • GitHub Repo: **https://github.com/ibaibhav0/ecoloop**

OFFICIAL ATTESTATION & PRODUCTION DISCLOSURE:
This technical document represents the complete operational and architectural specification of the EcoLoop platform deployed for **Nitte Deemed to be University**. All features described are fully active in production at **https://ecoloop-campus.vercel.app** and committed to the official repository at **https://github.com/ibaibhav0/ecoloop**.